

2 Description

The STM32C071x8/xB mainstream microcontrollers are based on high-performance Arm® Cortex®-M0+ 32-bit RISC core operating at up to 48 MHz frequency. Offering a high level of integration, they are suitable for a wide range of applications in consumer, industrial and appliance domains and ready for the Internet of Things (IoT) solutions.

The devices incorporate a memory protection unit (MPU), high-speed embedded memories (24 Kbytes of SRAM and up to 128 Kbytes of flash program memory with read and write protection, proprietary code protection, and securable area), DMA, an extensive range of system functions, enhanced I/Os, and peripherals. The devices offer standard communication interfaces (two I²Cs, two SPI / one I²S, one full-speed USB, and two USARTs), one 12-bit ADC (2.5 MSps) with up to 19 channels, a low-power RTC, an advanced control PWM timer, four general-purpose 16-bit timers, a 32-bit general-purpose timer, two watchdog timers, and a SysTick timer.

The devices operate within ambient temperatures from -40 to 125°C and with supply voltages from 2.0 V to 3.6 V. Optimized dynamic consumption combined with power-saving modes allows the design of low-power applications.

The devices are housed in packages with 19 to 64 pins.

Table 2. STM32C071x8/xB family device features and peripheral counts

Peripheral		STM32C071_									
		F8 / _FB	F8xxN / _FBxxN	G8 / _GB	G8xxN / _GBxxN	K8 / _KB	K8xxN / _KBxxN	C8 / _CB	C8xxN / _CBxxN	R8 / _RB	R8xxN / _RBxxN
Flash memory (Kbyte)		64/128 ⁽¹⁾	64/128								64/128 ⁽²⁾
SRAM (Kbyte)		24 with parity									
Timers	Advanced control	1 (16-bit)									
	General-purpose	1 (32-bit)									
		4 (16-bit)									
	SysTick	1									
	Watchdog	2									
Communication	SPI [I2S] ⁽³⁾	2 [1] + 2 extra through USARTs									
	I2C	2									
	USART	2									
	USB	1									
RTC		Yes									
VDDIO2 pin		No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
GPIOs (all 5V-tolerant)		18 or 17 ⁽⁴⁾	17	26	25	30	29	45	43	61	59
DMA channels		5									
Wakeup pins		4		5		5		5		6	